

Due on Tuesday April 12th, during class starting at noon
 Late submissions will be accepted till Friday April 15th morning 9AM, but
 since the solutions will be discussed in tutorial Friday 15th morning,
 submissions cannot be accepted after that.

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1. **[4+4 pts.]** Show that

$$\frac{d\gamma}{dt} = \frac{1}{c^2} \gamma^3 v \dot{v} \quad \text{and} \quad \frac{d\gamma}{dt} = \frac{1}{c^2} \gamma^3 \vec{v} \cdot \vec{a}$$

where we use the shorthands $\gamma = \gamma(v)$ and $\dot{v} = \frac{dv}{dt}$. Note that $\dot{v}$ is the derivative of the speed, not the derivative of the 3-vector $\vec{v}$, i.e., $\dot{v}$ is not the 3-acceleration $\vec{a} = \frac{d\vec{v}}{dt}$.

2. $A = (A_0, A_1, A_2, A_3)$ and $B = (B_0, B_1, B_2, B_3)$ are 4-vectors. Under a boost (Lorentz transformation), the components of A transform as

$$A'_0 = \gamma(v) \left(A_0 - \frac{v}{c} A_1 \right), \quad A'_1 = \gamma(v) \left(A_1 - \frac{v}{c} A_0 \right), \quad A'_2 = A_2, \quad A'_3 = A_3$$

- (a) **[2 pts.]** Write down the transformations for the components of B , under the same boost.
 (b) **[5 pts.]** Show that the inner product, defined as

$$A \cdot B = A_0 B_0 - A_1 B_1 - A_2 B_2 - A_3 B_3$$

is invariant under the same boost.

3. **[5 pts.]** The 4-velocity and 4-acceleration of a particle are given by

$$V = (\gamma c, \gamma \mathbf{v})$$

$$A = \left(\frac{\gamma^4}{c} v \dot{v}, \frac{\gamma^4}{c^2} v \dot{v} \mathbf{v} + \gamma^2 \mathbf{a} \right)$$

where we have used the shorthands $\gamma = \gamma(v)$ and $\dot{v} = \frac{dv}{dt}$, the speed $v = \sqrt{\mathbf{v} \cdot \mathbf{v}}$ is the magnitude of the 3-velocity $\mathbf{v}$, and $\mathbf{a}$ is the 3-acceleration. Show that the 4-velocity and 4-acceleration are ‘perpendicular’ to each other, i.e., have zero inner product.

4. In the lab frame, particle B moves to the right with speed u , and particle C moves to the left with speed v . In the frame of C , particle B is seen to move to the right with speed w , while particle C itself is of course at rest. Of course, w can be written down in terms of u and v using the velocity addition formula, but we will re-derive this formula below using 4-velocities. I encourage you to use $c = 1$ units for this problem. (But you can choose.) If you write γ , please specify which speed that γ corresponds to, e.g., through notation like $\gamma(u)$ or γ_u .
- (a) **[3 pts.]** In the lab frame, write down the 4-velocity for particle B and for particle C .
 - (b) **[3 pts.]** In C 's frame, write down the 4-velocity for particle B and for particle C .
 - (c) **[3 pts.]** The inner product of the two 4-velocities should be invariant, i.e., should be the same whichever frame you use. Write down an equation equating the inner product in the lab frame and in C 's frame.
 - (d) **[3 pts.]** From the inner product invariance, derive an expression for γ_w in terms of γ_u , γ_v , u and v .
 - (e) **[3 pts.]** From the equation relating γ 's, derive an expression for w in terms of u and v . You should have recovered the velocity addition formula.
5. **[2+3 pts.]** Two rockets P and Q approach each other on a collision course, moving (relative to the moon) at speeds $\frac{3}{5}c$ and $\frac{2}{5}c$ respectively. Find the speed of rocket Q as observed by an occupant of P .
- Relative to the moon, a third rocket R is seen to be chasing P (moving in the same direction as P) at speed $\frac{4}{5}c$. What is the speed of R as observed by the occupant of P ?